



```
=====
LOADING NEWS DATASET
=====
```

```
Dataset loaded successfully!
Total articles: 1015
```

```
Columns:
['category', 'title', 'body']
```

```
Missing values:
category    0
title       0
body        1
dtype: int64
```

```
Duplicate articles removed: 1
```

```
=====
NEWS CATEGORY DISTRIBUTION
=====
```

```
category
ARTS & CULTURE    1001
BUSINESS           13
Name: count, dtype: int64
```

```
=====
NLP PREPROCESSING
=====
```

```
Tokenization + Stop Word Removal + Stemming
Preprocessing completed!
```

```
=====
PREPROCESSING EXAMPLE
=====
```





```
ORIGINAL ARTICLE:
... Modeling Agencies Enabled Sexual Predators For Years, Former Agent Says In October 2017, Carolyn Kramer received a disturbing phone call. The former modeling agent listened

PROCESSED ARTICLE:
model agenc enabl sexual predat year former agent say octob carolyn kramer receiv disturb phone call former model agent listen intent model use repres told famou french phot

=====
      TRAIN TEST SPLIT
=====
Training articles: 811
Testing articles: 203

=====
      TF-IDF
=====
TF-IDF completed!
Number of TF-IDF features: 5000

=====
      TF-IDF SAMPLE TABLE
=====
Showing first 20 TF-IDF features:
```

	abaku	abandon	ab11	abl	abort	abraham	abramov	absolut	absorb	abstract	absurd	abt	abu	abus	academ	academi	accent	accept	access	accid
0	0.0	0.0000	0.0208	0.0	0.0	0.0	0.0	0.0	0.0	0.0491	0.0	0.0	0.0	0.0000	0.0	0.0	0.0	0.0214	0.0	0.0311
1	0.0	0.0000	0.0000	0.0	0.0	0.0	0.0	0.0	0.0	0.0000	0.0	0.0	0.0	0.0000	0.0	0.0	0.0	0.0000	0.0	0.0000
2	0.0	0.0000	0.0000	0.0	0.0	0.0	0.0	0.0	0.0	0.0000	0.0	0.0	0.0	0.0000	0.0	0.0	0.0	0.0715	0.0	0.0000
3	0.0	0.0218	0.0173	0.0	0.0	0.0	0.0	0.0	0.0	0.0000	0.0	0.0	0.0	0.0000	0.0	0.0	0.0	0.0000	0.0	0.0000
4	0.0	0.0000	0.0000	0.0	0.0	0.0	0.0	0.0	0.0	0.0571	0.0	0.0	0.0	0.0284	0.0	0.0	0.0	0.0000	0.0	0.0000





ir



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Commands + Code + Text ▶ Run all

RAM
Disk

=====

MODEL TRAINING

=====

Model training completed!

=====

MODEL ACCURACY

=====

Accuracy: 98.52 %

=====

CLASSIFICATION REPORT

=====

	precision	recall	f1-score	support
ARTS & CULTURE	0.99	1.00	0.99	200
BUSINESS	0.00	0.00	0.00	3
accuracy			0.99	203
macro avg	0.49	0.50	0.50	203
weighted avg	0.97	0.99	0.98	203

=====

NEWS NLP ANALYZER

=====

MODEL IS READY!

Enter any news-related query.

The program automatically performs:

Original Text

↓

Tokenization

!

{ Variables Terminal

Executing (6m 8s) Python 3



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Disk

Enter any news-related query.



*** The program automatically performs:

Original Text

↓

Tokenization

↓

Stop Word Identification

↓

Stop Word Removal

↓

Stemming

↓

TF-IDF

↓

News Category Prediction

Type 'exit' to stop.

Enter your query: A new smartphone company has launched an advanced artificial intelligence device with improved battery life and camera technology. The cricket team won 1

=====

NLP ANALYSIS

=====

1. ORIGINAL TEXT

A new smartphone company has launched an advanced artificial intelligence device with improved battery life and camera technology. The cricket team won the match after an

2. TOKENIZATION

['a', 'new', 'smartphone', 'company', 'has', 'launched', 'an', 'advanced', 'artificial', 'intelligence', 'device', 'with', 'improved', 'battery', 'life', 'and', 'camera', 't

3. STOP WORDS FOUND

{ Variables Terminal

Executing (7m 0s)

Python 3



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5. STEMMING



Original Word Stemmed Word

0	new	new
1	smartphone	smartphon
2	company	compani
3	launched	launch
4	advanced	advanc
5	artificial	artifici
6	intelligence	intellig
7	device	devic
8	improved	improv
9	battery	batteri
10	life	life
11	camera	camera
12	technology	technolog
13	cricket	cricket
14	team	team
15	match	match
16	outstanding	outstand

Variables Terminal



Executing (7m 52s) Python 3



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RAM



Disk



7. TF-IDF VALUES FOR QUERY



	Word	TF-IDF
287	artifici	0.3275
559	bowl	0.3221
3160	outstand	0.3083
1187	devic	0.2875
2186	improv	0.2697
2254	intellig	0.2676
2721	match	0.2525
63	advanc	0.2327
330	attack	0.2327
2458	launch	0.2258
647	camera	0.2227
4398	technolog	0.2168
4391	team	0.2012
4273	strong	0.1963
882	compani	0.1744
3276	perform	0.1335
2516	life	0.1117

Variables

Terminal



Executing (8m 28s)

Python 3



ir



File Edit View Insert Runtime Tools Help



Share



K

Commands + Code + Text Run all

RAM
Disk

647 camera 0.2227

4398 technolog 0.2168

4391 team 0.2012

4273 strong 0.1963

882 compani 0.1744

3276 perform 0.1335

2516 life 0.1117

2975 new 0.0870

8. TOP 3 PREDICTED CATEGORIES

Category Probability (%)

0 ARTS & CULTURE 98.42

1 BUSINESS 1.58

9. FINAL PREDICTION

Predicted Category: ARTS & CULTURE

Confidence: 98.42 %

ANALYSIS COMPLETE

Enter your query:

Variables

Terminal

Executing (8m 50s)

Python 3